

2(a)	<i>any one from:</i> frog, age / size / mass / stage of maturity / from same population ; light intensity ; humidity ; time of day ; time of year ; duration of recording ; background noise ;	1
2(b)	as (body) temperature increases the interval between calls decreases ;	1
2(c)(i)	<i>any three from:</i> 1 data (for temperature and calls) is, paired / linked ; 2 <u>data</u> is continuous ; 3 <u>data</u> shows normal distribution ; 4 scatter diagram, suggests a linear, relationship / correlation ; 5 there are at least five (paired) observations ;	3
2(c)(ii)	there is no, correlation / relationship, between the intervals between calls and temperature ;	1
2(c)(iii)	<i>any two from:</i> <i>significant because</i> 1 (critical value for $p = 0.05 / 5\%$ is) 0.279 or (critical value for $p = 0.01 / 1\%$ is) 0.361 ; 2 value for $r / 0.717$, is $>$ than critical value (so null hypothesis is rejected / accept the alternative hypothesis) / ora ;	2

Question	Answer	Marks
2(d)	any two from: 1 duration of each call ; 2 number of calls in a set time ; 3 number / frequency, of pulses, within each call ; 4 length of time between pulses within each call ; 5 <u>sound</u> frequency (kHz) / pitch ; 6 duration of each pulse ;	2
2(e)	any three from: 1 capture frogs and count / find number / record number and mark and release them ; 2 suitable marking so not harmful / too obvious / not removed or appropriate time between first and second capture so frogs time to mix ; 3 capture second sample and count / find number / record number of frogs and count / find number / record number, of marked frogs ; 4 use Lincoln / Petersen, index ;	3